

CALCULUS WITH

Asante

TOPIC THREE | TOPIC SERIES

Derivatives

Compiled and taught by Asante

MATH 152 | Calculus & Analysis | KNUST BME1

Why This Booklet Exists

This booklet is here to supplement your own learning — not replace it. Lectures, your notes, your textbook, and the time you spend wrestling with a problem on your own all still matter. Nothing here is a shortcut around doing the work; it's a second angle on the same material, laid out the way I wish someone had laid it out for me.

Learning is not a straight line. You will understand the Chain Rule on page 11, forget it exists two weeks later, and only really get it when it shows up again in a totally different problem. That's normal. This booklet is built to be revisited, not read once and shelved.

If you get stuck, that's the point where the real learning starts, *chale* — not a sign that something is wrong with you. Sit with the confusion for a bit before you flip to the answer.

Every booklet in this series also closes with a Cheat Sheet — a fast-reference summary of the whole topic. It's there for review the night before a test, not for learning something for the first time.

A Note on How to Use This

Don't try to read this cover to cover in one sitting the night before a test. Work through one section, attempt the boxed questions honestly, and come back to whatever didn't stick.

How This Booklet Works

Every topic in this series is laid out the same way, so once you learn the pattern once, you can move faster through every booklet after this one. Here's what each marker means.

TRY THIS FIRST

A few questions up front, before any explanation. Attempt them cold — they tell you exactly which part of this page you actually need.

DEFINITION

The formal statement — rule, theorem, or term — stated precisely. Worth memorising word for word.

EXAMPLE

A worked problem, solved in full, showing the reasoning step by step, not just the answer.

TRY IT

Your turn. Attempt it before checking the answer given underneath.

RECALL / STRATEGY

A shortcut, common trap, or the exact test to reach for — the thing you should be thinking right before you attempt a problem.

WATCH & LEARN

Hyperlinked videos, dropped in right next to the concept they explain — click (▶) whenever reading alone isn't landing.

Each booklet ends with a Problem Set and a Cheat Sheet summary. Questions

marked **T** are technology-appropriate — a calculator or CAS is fine for checking arithmetic, not for skipping the reasoning. Full worked solutions live in the separate Answer Key that accompanies this booklet.

Asante Inc. understands that learning is not a linear process and can be stressful — so this booklet has hyperlinked videos built in throughout, plus a short list of videos worth watching before you even start reading. Look for ► next to any concept that isn't landing.

Key Vocabulary

A quick glossary before you dive in. Come back here any time a term stops making sense mid-page.

Derivative

The instantaneous rate of change of a function — like a car's speedometer reading at one exact instant, not the average speed over the whole trip.

Differentiability

A function is differentiable where it has one clearly defined tangent slope. A smooth road has one direction at every point; a sharp corner doesn't.

Tangent Line

The line that just grazes a curve at one point, matching its direction there — like headlights pointing exactly where the road curves at this instant.

Chain Rule

The rule for differentiating a function nested inside another — like tracking how fast a shadow moves when both the sun's angle and your walking speed are changing at once.

Key Vocabulary (continued)

Implicit Function

A relationship between x and y that isn't solved for y — like the equation of a circle, where y isn't given directly but is still tied to x .

n -th Derivative

The result of differentiating n times in a row. The first derivative is speed, the second is acceleration, the third tracks how quickly the acceleration itself is changing.

L'Hôpital's Rule

A rule for resolving a limit where numerator and denominator are “tied” (both $\rightarrow 0$ or both $\rightarrow \infty$), by comparing their rates of change instead of their raw values.

Recommended Videos --- Watch Before Reading

Just these five before you start — they carry the backbone ideas the rest of the booklet builds on.

- ▶ [Differentiation from First Principles](#)
- ▶ [Derivatives of Elementary Functions](#)
- ▶ [Differentiation of Composite Functions — The Chain Rule](#)
- ▶ [Implicit Differentiation](#)
- ▶ [L'Hôpital's Rule — Indeterminate Quotients](#)

Every video used across this booklet lives in one playlist:

- ▶ [Full Playlist — Differentiation, SkanCity Academy](#)

Bonus: [Professor Leonard — full-length lecture](#) — not tied to one section, but worth the time if you want the deeper, slower-paced version of this whole topic.

Try This First --- 5 Questions, No Notes

1. Use first principles to find $f'(x)$ for $f(x) = x^2$.

- (A) x
- (B) 2
- (C) $2x$
- (D) x^2

2. Differentiate $f(x) = \sin(x^2)$.

- (A) $2x \cos(x^2)$
- (B) $\cos(x^2)$
- (C) $2x \sin(x^2)$
- (D) $-2x \sin(x^2)$

3. Differentiate $f(x) = x^2 \sin x$.

- (A) $2x \sin x + x^2 \cos x$
- (B) $2x \cos x + x^2 \sin x$
- (C) $x^2 \cos x$
- (D) $2x \sin x - x^2 \cos x$

4. If $x^2 + y^2 = 25$, find $\frac{dy}{dx}$.

- (A) $\frac{x}{y}$
- (B) $-\frac{x}{y}$
- (C) $-\frac{y}{x}$
- (D) $\frac{y}{x}$

5. Find $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$.

- (A) 0
- (B) 1
- (C) ∞
- (D) Does not exist

HOW DID THAT FEEL?

Questions 2 and 3 look similar but need completely different rules — mixing them up is the single most common differentiation mistake. If you paused on either, that's exactly what this booklet sorts out.

1. Differentiation from First Principles

DEFINITION --- DERIVATIVE FROM FIRST PRINCIPLES

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Every differentiation rule in this booklet is just a shortcut for this one limit, worked out in advance for common function shapes.

Example 1

Use first principles to differentiate $f(x) = x^2$.

Solution.

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x$$

RECALL / STRATEGY --- INTERVAL OF DIFFERENTIABILITY

A function is only differentiable where the first-principles limit actually exists. $f(x) = \sqrt{x-1}$ has domain $x \geq 1$, but the derivative $\frac{1}{2\sqrt{x-1}}$ blows up at $x = 1$ itself — so f is differentiable only on the open interval $x > 1$, not the closed domain.

► Watch: Differentiation from First Principles

2. Derivatives of Elementary Functions

DEFINITION --- STANDARD DERIVATIVES

$$\begin{array}{lll} \frac{d}{dx} x^n = nx^{n-1} & \frac{d}{dx} \sin x = \cos x & \frac{d}{dx} \cos x = -\sin x \\ \frac{d}{dx} e^x = e^x & \frac{d}{dx} \ln x = \frac{1}{x} & \end{array}$$

Example 2

Differentiate $f(x) = 3x^4 - 5x^2 + 7$.

Solution. Apply the power rule term by term: $f'(x) = 12x^3 - 10x$.

► [Watch: Derivatives of Elementary Functions](#)

3. Derivatives of Inverse Trig Functions

DEFINITION --- INVERSE TRIG DERIVATIVES

$$\begin{array}{lll} \frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}} & \frac{d}{dx} \cos^{-1} x = -\frac{1}{\sqrt{1-x^2}} & \frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2} \end{array}$$

Example 3

Differentiate $f(x) = \tan^{-1}(3x)$.

Solution. Combine the $\tan^{-1}$ derivative with the Chain Rule (Section 5): $f'(x) =$

$$\frac{1}{1+(3x)^2} \cdot 3 = \frac{3}{1+9x^2}.$$

► [Watch: Derivatives of Inverse Trigonometric Functions](#)

4. Derivatives of Hyperbolic Functions

DEFINITION --- HYPERBOLIC DERIVATIVES

$$\frac{d}{dx} \sinh x = \cosh x \quad \frac{d}{dx} \cosh x = \sinh x \quad \frac{d}{dx} \tanh x = \operatorname{sech}^2 x$$

Example 4

Differentiate $f(x) = \cosh(2x)$.

Solution. Combine the hyperbolic derivative with the Chain Rule: $f'(x) = \sinh(2x) \cdot 2 = 2 \sinh(2x)$.

► **Watch:** Derivatives of Hyperbolic and Inverse Hyperbolic Functions

5. Differentiation of Composite Functions --- The Chain Rule

DEFINITION --- CHAIN RULE

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

Differentiate the outside function first, leaving the inside untouched, then multiply by the derivative of the inside.

Example 5

Differentiate $f(x) = \sin(x^2)$.

Solution. Outside function $\sin(\cdot)$, inside function x^2 :

$$f'(x) = \cos(x^2) \cdot \frac{d}{dx}(x^2) = 2x \cos(x^2)$$

► **Watch:** Derivative of Composite Functions --- The Chain Rule

6. Differentiation of Product of Functions --- The Product Rule

DEFINITION --- PRODUCT RULE

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$$

RECALL / STRATEGY --- DON'T JUST MULTIPLY THE DERIVATIVES

$(fg)' \neq f'g'$. This is the single most common error with this rule — both terms in the sum matter.

Example 6

Differentiate $f(x) = x^2 \sin x$.

Solution. Let $u = x^2$ ($u' = 2x$) and $v = \sin x$ ($v' = \cos x$):

$$f'(x) = 2x \sin x + x^2 \cos x$$

► Watch: Differentiation of Product of Functions --- The Product Rule

7. Differentiation of Rational Functions --- The Quotient Rule

DEFINITION --- QUOTIENT RULE

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Example 7

Differentiate $f(x) = \frac{x^2 + 1}{x - 1}$.

Solution. Let $u = x^2 + 1$ ($u' = 2x$) and $v = x - 1$ ($v' = 1$):

$$f'(x) = \frac{2x(x - 1) - (x^2 + 1)(1)}{(x - 1)^2} = \frac{x^2 - 2x - 1}{(x - 1)^2}$$

TRY IT --- YOUR TURN

Differentiate $f(x) = \frac{\sin x}{x}$.

Answer: $f'(x) = \frac{x \cos x - \sin x}{x^2}$

► Watch: Differentiation of Rational Functions --- The Quotient Rule

8. Implicit Differentiation

RECALL / STRATEGY --- EVERY y PICKS UP A dy/dx

When x and y are tangled together in one equation, differentiate both sides with respect to x . Every time you differentiate a y -term, the Chain Rule attaches a $\frac{dy}{dx}$ to it — that's the whole trick.

Example 8

Find $\frac{dy}{dx}$ if $x^2 + y^2 = 25$.

Solution. Differentiate both sides with respect to x :

$$2x + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}$$

► Watch: Implicit Differentiation

9. Leibnitz Theorem --- Higher Order Derivatives

DEFINITION --- LEIBNITZ'S THEOREM

For a product $f(x)g(x)$, the n -th derivative is

$$(fg)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n-k)}$$

A generalisation of the Product Rule to any order, not just the first derivative.

Example 9

Find the third derivative of $f(x) = x^2 e^x$ using Leibnitz's Theorem.

Solution. With $u = x^2$ ($u' = 2x$, $u'' = 2$, $u''' = 0$) and $v = e^x$ (every derivative is e^x):

$$(uv)''' = u'''v + 3u''v' + 3u'v'' + uv''' = 0 + 3(2)e^x + 3(2x)e^x + x^2e^x = (x^2 + 6x + 6)e^x$$

► Watch: Leibnitz Theorem --- Higher Order Derivatives

10. L'Hôpital's Rule --- Indeterminate Quotients

DEFINITION --- L'HÔPITAL'S RULE

If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

provided the right-hand limit exists. Differentiate top and bottom separately — never use the Quotient Rule here.

Example 10

Find $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$.

Solution. This is $\frac{\infty}{\infty}$. Differentiate top and bottom separately:

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

► Watch: L'Hôpital's Rule --- Indeterminate Quotients

11. L'Hôpital's Rule --- Indeterminate Products

RECALL / STRATEGY --- TURN A PRODUCT INTO A QUOTIENT

$0 \cdot \infty$ isn't directly usable by L'Hôpital's Rule. Rewrite one factor as $1/(\text{its reciprocal})$ to force a $\frac{0}{0}$ or $\frac{\infty}{\infty}$ shape, then apply the rule as usual.

Example 11

Find $\lim_{x \rightarrow 0^+} x \ln x$.

Solution. This is $0 \cdot (-\infty)$. Rewrite $x \ln x = \frac{\ln x}{1/x}$, which is now $\frac{-\infty}{\infty}$:

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0$$

► Watch: L'Hôpital's Rule --- Indeterminate Products

12. L'Hôpital's Rule --- Indeterminate Powers

RECALL / STRATEGY --- TAKE THE LOG FIRST

Forms like 0^0 , 1^∞ , and ∞^0 need a log transform before L'Hôpital's Rule applies. Set y equal to the expression, take $\ln y$ on both sides, solve that limit, then exponentiate back at the end.

Example 12

Find $\lim_{x \rightarrow 0^+} x^x$.

Solution. This is 0^0 . Let $y = x^x$, so $\ln y = x \ln x$. From Example 11, $\lim_{x \rightarrow 0^+} x \ln x = 0$, so

$$\lim_{x \rightarrow 0^+} \ln y = 0 \implies \lim_{x \rightarrow 0^+} y = e^0 = 1$$

► Watch: L'Hôpital's Rule --- Indeterminate Powers

Putting It Together

- Every rule in this booklet is a pre-solved shortcut for the first-principles limit.
- Differentiability needs an open interval — check the endpoints of a domain separately.
- Chain Rule: outside derivative, inside untouched, times the inside's derivative.
- Product Rule: $f'g + fg'$ — never just $f'g'$. Quotient Rule: $\frac{f'g - fg'}{g^2}$.
- Implicit differentiation: every y -term picks up a $\frac{dy}{dx}$ via the Chain Rule.
- Leibnitz's Theorem generalises the Product Rule to any derivative order using binomial coefficients.
- L'Hôpital's Rule only applies directly to $\frac{0}{0}$ or $\frac{\infty}{\infty}$: rewrite products as quotients, and take logs first for indeterminate powers.

Problem Set

Questions marked **T** are technology-appropriate — a calculator or CAS is fine for checking your arithmetic, not for skipping the reasoning.

1. Use first principles to find $f'(x)$ for $f(x) = x^3$.

(A) $3x^2$

(B) x^3

(C) $3x$

(D) x^2

2. Differentiate $f(x) = \cos(5x^2)$.

(A) $-5x \sin(5x^2)$

(B) $-10x \sin(5x^2)$

(C) $10x \sin(5x^2)$

(D) $-10x \cos(5x^2)$

3. Differentiate $f(x) = x^3 \cos x$.

(A) $3x^2 \cos x + x^3 \sin x$

(B) $x^3 \sin x$

(C) $3x^2 \cos x - x^3 \sin x$

(D) $3x^2 \sin x - x^3 \cos x$

4. Differentiate $f(x) = \frac{x}{x^2 + 1}$.

(A) $\frac{1 + x^2}{(x^2 + 1)^2}$

(B) $\frac{1}{(x^2 + 1)^2}$

(C) $\frac{x^2 - 1}{(x^2 + 1)^2}$

(D) $\frac{1 - x^2}{(x^2 + 1)^2}$

5. Differentiate $f(x) = \sin^{-1}(2x)$.

(A) $\frac{1}{\sqrt{1-4x^2}}$

(B) $\frac{2}{\sqrt{1-4x^2}}$

(C) $\frac{2}{\sqrt{1-x^2}}$

(D) $\frac{1}{\sqrt{1-2x^2}}$

6. If $x^3 + y^3 = 6xy$, find $\frac{dy}{dx}$.

(A) $\frac{2y - x^2}{y^2 - 2x}$

(B) $\frac{x^2 - 2y}{y^2 - 2x}$

(C) $\frac{2y - x^2}{y^2}$

(D) $\frac{x^2}{y^2}$

7. **T** Find the second derivative of $f(x) = x^3 \ln x$ using Leibnitz's Theorem.

(A) $6x \ln x + 6x$

(B) $3x \ln x + 5x$

(C) $6x \ln x + 5x$

(D) $6x \ln x - 5x$

8. Find $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$.

(A) 0

(B) e

(C) ∞

(D) 1

9. Find $\lim_{x \rightarrow \infty} x \sin(1/x)$.

(A) 0

(B) 1

(C) ∞

(D) Does not exist

10. **T** Find $\lim_{x \rightarrow 0^+} (1 + x)^{1/x}$.

(A) 1

(B) ∞

(C) 0

(D) e

Cheat Sheet

The pages that follow are Paul Dawkins's widely used Calculus Cheat Sheet, included here as a fast-reference summary — not a replacement for working through this booklet. All credit for this reference sheet belongs to its original author.

Derivatives

Basic Properties/Formulas/Rules

$$\frac{d}{dx}(cf(x)) = cf'(x), \text{ c is any constant.}$$

$$\frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$$

$$\frac{d}{dx}(x^n) = nx^{n-1}, \text{ n is any number.}$$

$$\frac{d}{dx}(c) = 0, \text{ c is any constant.}$$

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x) - \textbf{Product Rule}$$

$$\frac{d}{dx}(e^{g(x)}) = g'(x)e^{g(x)}$$

$$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} - \textbf{Quotient Rule}$$

$$\frac{d}{dx}[\ln(g(x))] = \frac{g'(x)}{g(x)}$$

$$\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x) - \textbf{Chain Rule}$$

Common Derivatives

Polynomials

$$\frac{d}{dx}(c) = 0$$

$$\frac{d}{dx}(x) = 1$$

$$\frac{d}{dx}(cx) = c$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(cx^n) = ncx^{n-1}$$

Trig Functions

$$\frac{d}{dx}[\sin(x)] = \cos(x)$$

$$\frac{d}{dx}[\cos(x)] = -\sin(x)$$

$$\frac{d}{dx}[\tan(x)] = \sec^2(x)$$

$$\frac{d}{dx}[\csc(x)] = -\csc(x)\cot(x)$$

$$\frac{d}{dx}[\sec(x)] = \sec(x)\tan(x)$$

$$\frac{d}{dx}[\cot(x)] = -\csc^2(x)$$

Inverse Trig Functions

$$\frac{d}{dx}[\sin^{-1}(x)] = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}[\cos^{-1}(x)] = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}[\tan^{-1}(x)] = \frac{1}{1+x^2}$$

$$\frac{d}{dx}[\csc^{-1}(x)] = -\frac{1}{|x|\sqrt{x^2-1}}$$

$$\frac{d}{dx}[\sec^{-1}(x)] = \frac{1}{|x|\sqrt{x^2-1}}$$

$$\frac{d}{dx}[\cot^{-1}(x)] = -\frac{1}{1+x^2}$$

Exponential & Logarithm Functions

$$\frac{d}{dx}[a^x] = a^x \ln(a)$$

$$\frac{d}{dx}[e^x] = e^x$$

$$\frac{d}{dx}[\ln(x)] = \frac{1}{x}, \quad x > 0$$

$$\frac{d}{dx}[\ln|x|] = \frac{1}{x}, \quad x \neq 0$$

$$\frac{d}{dx}[\log_a(x)] = \frac{1}{x \ln(a)}, \quad x > 0$$

Hyperbolic Functions

$$\frac{d}{dx}[\sinh(x)] = \cosh(x)$$

$$\frac{d}{dx}[\cosh(x)] = \sinh(x)$$

$$\frac{d}{dx}[\tanh(x)] = \text{sech}^2(x)$$

$$\frac{d}{dx}[\text{csch}(x)] = -\text{csch}(x)\coth(x)$$

$$\frac{d}{dx}[\text{sech}(x)] = -\text{sech}(x)\tanh(x)$$

$$\frac{d}{dx}[\coth(x)] = -\text{csch}^2(x)$$

Integrals

Basic Properties/Formulas/Rules

$$\int cf(x) dx = c \int f(x) dx, c \text{ is a constant.} \quad \int f(x) \pm g(x) dx = \int f(x) dx \pm \int g(x) dx$$

$$\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a) \text{ where } F(x) = \int f(x) dx$$

$$\int_a^b cf(x) dx = c \int_a^b f(x) dx, c \text{ is a constant.} \quad \int_a^b f(x) \pm g(x) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$$

$$\int_a^a f(x) dx = 0$$

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx \quad \int_a^b c dx = c(b-a), c \text{ is a constant.}$$

$$\text{If } f(x) \geq 0 \text{ on } a \leq x \leq b \text{ then } \int_a^b f(x) dx \geq 0$$

$$\text{If } f(x) \geq g(x) \text{ on } a \leq x \leq b \text{ then } \int_a^b f(x) dx \geq \int_a^b g(x) dx$$

Common Integrals

Polynomials

$$\int dx = x + c \quad \int k dx = kx + c \quad \int x^n dx = \frac{1}{n+1} x^{n+1} + c, n \neq -1$$

$$\int \frac{1}{x} dx = \ln|x| + c \quad \int x^{-1} dx = \ln|x| + c \quad \int x^{-n} dx = \frac{1}{-n+1} x^{-n+1} + c, n \neq 1$$

$$\int \frac{1}{ax+b} dx = \frac{1}{a} \ln|ax+b| + c \quad \int x^{\frac{p}{q}} dx = \frac{1}{\frac{p}{q}+1} x^{\frac{p}{q}+1} + c = \frac{q}{p+q} x^{\frac{p+q}{q}} + c$$

Trig Functions

$$\int \cos(u) du = \sin(u) + c \quad \int \sin(u) du = -\cos(u) + c \quad \int \sec^2 u du = \tan(u) + c$$

$$\int \sec(u) \tan(u) du = \sec(u) + c \quad \int \csc(u) \cot(u) du = -\csc(u) + c \quad \int \csc^2 u du = -\cot(u) + c$$

$$\int \tan(u) du = -\ln|\cos(u)| + c = \ln|\sec(u)| + c \quad \int \cot(u) du = \ln|\sin(u)| + c = -\ln|\csc(u)| + c$$

$$\int \sec(u) du = \ln|\sec(u) + \tan(u)| + c \quad \int \sec^3(u) du = \frac{1}{2} \left(\sec(u) \tan(u) + \ln|\sec(u) + \tan(u)| \right) + c$$

$$\int \csc(u) du = \ln|\csc(u) - \cot(u)| + c \quad \int \csc^3(u) du = \frac{1}{2} \left(-\csc(u) \cot(u) + \ln|\csc(u) - \cot(u)| \right) + c$$

Exponential & Logarithm Functions

$$\int e^u du = e^u + c \quad \int a^u du = \frac{a^u}{\ln(a)} + c \quad \int \ln(u) du = u \ln(u) - u + c$$

$$\int e^{au} \sin(bu) du = \frac{e^{au}}{a^2 + b^2} (a \sin(bu) - b \cos(bu)) + c \quad \int u e^u du = (u - 1)e^u + c$$

$$\int e^{au} \cos(bu) du = \frac{e^{au}}{a^2 + b^2} (a \cos(bu) + b \sin(bu)) + c \quad \int \frac{1}{u \ln(u)} du = \ln |\ln(u)| + c$$

Inverse Trig Functions

$$\int \frac{1}{\sqrt{a^2 - u^2}} du = \sin^{-1} \left(\frac{u}{a} \right) + c \quad \int \sin^{-1}(u) du = u \sin^{-1}(u) + \sqrt{1 - u^2} + c$$

$$\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + c \quad \int \tan^{-1}(u) du = u \tan^{-1}(u) - \frac{1}{2} \ln(1 + u^2) + c$$

$$\int \frac{1}{u \sqrt{u^2 - a^2}} du = \frac{1}{a} \sec^{-1} \left(\frac{u}{a} \right) + c \quad \int \cos^{-1}(u) du = u \cos^{-1}(u) - \sqrt{1 - u^2} + c$$

Hyperbolic Functions

$$\int \sinh(u) du = \cosh(u) + c \quad \int \operatorname{sech}(u) \tanh(u) du = -\operatorname{sech}(u) + c \quad \int \operatorname{sech}^2(u) du = \tanh(u) + c$$

$$\int \cosh(u) du = \sinh(u) + c \quad \int \operatorname{csch}(u) \coth(u) du = -\operatorname{csch}(u) + c \quad \int \operatorname{csch}^2(u) du = -\coth(u) + c$$

$$\int \tanh(u) du = \ln(\cosh(u)) + c \quad \int \operatorname{sech}(u) du = \tan^{-1} |\sinh(u)| + c$$

Miscellaneous

$$\int \frac{1}{a^2 - u^2} du = \frac{1}{2a} \ln \left| \frac{u+a}{u-a} \right| + c \quad \int \sqrt{a^2 + u^2} du = \frac{u}{2} \sqrt{a^2 + u^2} + \frac{a^2}{2} \ln |u + \sqrt{a^2 + u^2}| + c$$

$$\int \frac{1}{u^2 - a^2} du = \frac{1}{2a} \ln \left| \frac{u-a}{u+a} \right| + c \quad \int \sqrt{u^2 - a^2} du = \frac{u}{2} \sqrt{u^2 - a^2} - \frac{a^2}{2} \ln |u + \sqrt{u^2 - a^2}| + c$$

$$\int \sqrt{a^2 - u^2} du = \frac{u}{2} \sqrt{a^2 - u^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{u}{a} \right) + c$$

$$\int \sqrt{2au - u^2} du = \frac{u-a}{2} \sqrt{2au - u^2} + \frac{a^2}{2} \cos^{-1} \left(\frac{a-u}{a} \right) + c$$

Standard Integration Techniques

u Substitution: $\int_a^b f(g(x)) g'(x) dx$ will convert the integral into $\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$ using the substitution $u = g(x)$ where $du = g'(x) dx$. For indefinite integrals drop the limits of integration.

Integration by Parts : $\int u dv = uv - \int v du$ and $\int_a^b u dv = uv \Big|_a^b - \int_a^b v du$. Choose u and dv from integral and compute du by differentiating u and compute v using $v = \int dv$.

Trig Substitutions : If the integral contains the following root use the given substitution and formula.

$$\sqrt{a^2 - b^2 x^2} \Rightarrow x = \frac{a}{b} \sin(\theta) \quad \text{and} \quad \cos^2(\theta) = 1 - \sin^2(\theta)$$

$$\sqrt{b^2 x^2 - a^2} \Rightarrow x = \frac{a}{b} \sec(\theta) \quad \text{and} \quad \tan^2(\theta) = \sec^2(\theta) - 1$$

$$\sqrt{a^2 + b^2 x^2} \Rightarrow x = \frac{a}{b} \tan(\theta) \quad \text{and} \quad \sec^2(\theta) = 1 + \tan^2(\theta)$$

Partial Fractions : If integrating a rational expression involving polynomials, $\int \frac{P(x)}{Q(x)} dx$, where the degree (largest exponent) of $P(x)$ is smaller than the degree of $Q(x)$ then factor the denominator as completely as possible and find the partial fraction decomposition of the rational expression. Integrate the partial fraction decomposition (P.F.D.). For each factor in the denominator we get term(s) in the decomposition according to the following table.

Factor of $Q(x)$	Term in P.F.D	Factor is $Q(x)$	Term in P.F.D
$ax + b$	$\frac{A}{ax + b}$	$(ax + b)^k$	$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \dots + \frac{A_k}{(ax + b)^k}$
$ax^2 + bx + c$	$\frac{Ax + B}{ax^2 + bx + c}$	$(ax^2 + bx + c)^k$	$\frac{A_1 x + B_1}{ax^2 + bx + c} + \dots + \frac{A_k x + B_k}{(ax^2 + bx + c)^k}$

Products and (some) Quotients of Trig Functions :

For $\int \sin^n(x) \cos^m(x) dx$ we have the following :

1. **n odd.** Strip 1 sine out and convert rest to cosines using $\sin^2(x) = 1 - \cos^2(x)$, then use the substitution $u = \cos(x)$.
2. **m odd.** Strip 1 cosine out and convert rest to sines using $\cos^2(x) = 1 - \sin^2(x)$, then use the substitution $u = \sin(x)$.
3. **n and m both odd.** Use either 1. or 2.
4. **n and m both even.** Use double angle and/or half angle formulas to reduce the integral into a form that can be integrated.

For $\int \tan^n(x) \sec^m(x) dx$ we have the following :

1. **n odd.** Strip 1 tangent and 1 secant out and convert the rest to secants using $\tan^2(x) = \sec^2(x) - 1$, then use the substitution $u = \sec(x)$.
2. **m even.** Strip 2 secants out and convert rest to tangents using $\sec^2(x) = 1 + \tan^2(x)$, then use the substitution $u = \tan(x)$.
3. **n odd and m even.** Use either 1. or 2.
4. **n even and m odd.** Each integral will be dealt with differently.

Convert Example : $\cos^6(x) = (\cos^2(x))^3 = (1 - \sin^2(x))^3$